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WBD Discovery - Anti Piracy Strategy

June 2026 - Version 1.0

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Executive Summary

Warner Bros. Discovery (WBD) holds premium live football rights in Chile and Argentina, but the commercial value of those rights is eroding as audiences migrate from legitimate subscriptions to an increasingly sophisticated piracy ecosystem. In **Chile**, TNT Sports has fallen from roughly one million subscribers to 600,000 over five years, with the **~400,000 lost customers** most likely absorbed by piracy. The structural backdrop accelerates the trend: pay-TV households dropped from 3.5M to 2.5M while high speed internet grew to 4.8M houses, and Chilean internet is among the fastest in the world (600-800 Mbps), meaning illegal streams now run flawlessly. A pricing gap compounds the problem.

In this context**, Piracy** has matured into an **industrialized, resilient supply chain** feeding **three consumer tiers**: free **web streams** (the highest-traffic tier, with single domains drawing up to 8.8M monthly visits), paid **IPTV applications**, and preloaded **TV boxes**. IPTV, led by MagisTV (now rebranded SuperTV/Flujo TV), is the dominant and most damaging tier, accounting for roughly 20% of ISP network traffic and an estimated one-million-plus Chilean households, at about a third of the price of a legal bundle but with thousands of channels. Pirate demand splits into **two clear profiles**: the casual web-stream user, who watches for free and is unlikely to ever pay, and the IPTV user, who already pays for a cable-like service and is more likely to convert into a legitimate subscriber. IPTV is therefore the higher-value target.

Against this, **WBD's current anti-piracy operation is capable but fragmented**, heavily manual, and without a central data layer: valuable intelligence from premium vendors sits siloed across spreadsheets and email, with no single source of truth and no reliable baseline to measure impact. Official enforcement cannot keep pace with live sports: DMCA takedowns average around 59 minutes, close to an entire half, and real-time dynamic domain blocking is legally unavailable in both Chile and Argentina.

In this context, we recommend shifting from chasing individual live streams to disrupting the pirate ecosystem as a whole, through a modular three-layer strategy:

- **Layer 1 -** Data foundation: unify existing vendor intelligence into a single source of truth.
- **Layer 2** - Full-scope intelligence: map the entire pirate lifecycle, from social-media promotion through to payment collection.
- **Layer 3** - Automated disruption: target the pirates' real operational weak points, namely regulated payment rails, sales and communication channels, and, for web streams, advertising revenue.

To materialize this strategy, **three proof-of-concepts** will be implemented: a Data Layer Agent, a Social Media Monitor, and a disruption agent called Trinity. Built to be **market-agnostic**, they are designed to feed evidence to Legal and impose genuine operational cost on illicit networks. Concretely, the POCs target measurable outcomes: automated ingestion of ≥90% of vendor reports with time-to-insight cut from hours to minutes, a ≥3-5x increase in pirate sources discovered through social media at ≥80% detection precision, and measurable operational disruption of targeted IPTV services. The result is twofold: a complete, real-time picture of the pirate ecosystem, and the means to strike at its core and convert disruption into recovered subscribers.

These three PoCs will be developed following short, continuous delivery cycles in which the Marvik and WBD teams iterate together, ensuring the solution generates value from day one. Critically, they are not isolated implementations. They represent an integrated effort designed to evolve into production-grade systems, gradually building a robust antipiracy infrastructure within Warner Bros. Discovery.

Context & Methodology

General Context

The commercial value of Warner Bros. Discovery's premium live football rights in Chile and Argentina is facing a critical inflection point due to a rapidly maturing, industrialized piracy ecosystem. For instance, in Chile alone, premium sports subscriptions have decreased from approximately one million to 600,000 over a five-year period, resulting in a severe erosion of legitimate revenue. This shift is heavily accelerated by structural factors, including massive regional adoption of high-speed fiber internet and premium pricing tiers that leave cost-sensitive segments underserved.

Modern piracy has evolved far beyond casual web links into sophisticated, subscription-based IPTV networks that now account for nearly 20% of internet provider network traffic. This document establishes a business-driven strategy to move beyond standard, reactive stream-chasing. By shifting our focus toward systematically disrupting the economic and operational foundations of these illicit networks, WBD aims to reclaim market share, protect the value of our media rights, and directly convert disrupted users into legitimate subscribers.

Work Methodology

To formulate a highly targeted, data-backed strategy, a rigorous 4-week discovery process was executed, focused on auditing internal workflows, evaluating vendor's telemetry, and mapping pirate operational vulnerabilities. This cross-functional methodology consisted of:

- **Stakeholder Workshops**: Conducted extensive collaborative sessions across all internal divisions involved in the rights lifecycle including Anti-Piracy Management, Panregional and Local Legal, Digital Events Operations, the TNT Sports Chile Social Media team, and Regional Finance and Distribution.
- **Data Auditing**: Evaluated Latin American pirate consumer profiles, underlying network traffic demands, and the technical visibility provided by our current premium security vendors.
- **Operational Testing**: Audited the baseline capabilities of current signal security infrastructure, mapped the inefficiencies of existing manual workflows, and analyzed historical tactical successes to extract scalable blueprints for future deterrence

Document Structure & Roadmap

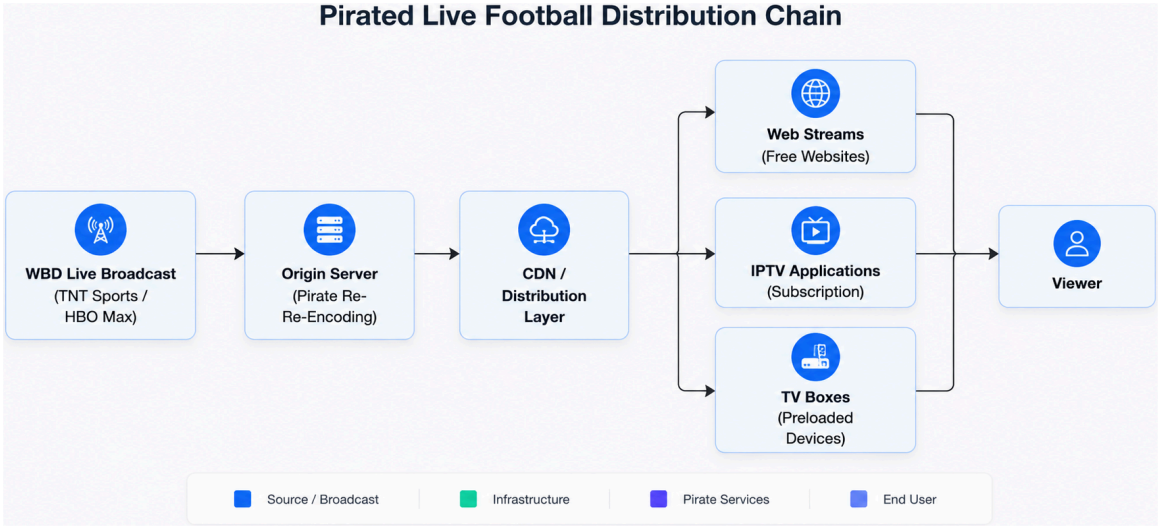
This strategic proposal is organized into five sequential chapters:

- **Sports Piracy Landscape**: An overview of the modern pirate distribution chain and a review of regional benchmark technologies.
- **Warner Bros. Discovery Assessment**: A baseline operational audit mapping out internal stakeholders, current constraints, and the limitations of our disconnected external provider matrix.
- **Strategic Diagnosis**: A critical analysis of WBD's structural vulnerabilities (such as data fragmentation and manual bottlenecks) alongside identified pirate weak points and core user conversion hypotheses.
- **Antipiracy Implementation Path**: A modular, three-layered strategic framework detailing the creation of a centralized data foundation, full-scope intelligence mapping, and automated operational disruption workflows.
- **Proofs of Concept (POCs) & Timeline**: Technical designs, operational KPIs, and execution timelines for three automated agents (Data Layer Agent, Social Media Monitor Agent, and Trinity Agent) designed to operationalize this strategy. Also, here the **economic and operational benefits** of this implementation are explained.

Read sequentially, these chapters form a single, coherent roadmap, with each building on the conclusions of the one before it. The result is not a set of isolated observations, but an actionable strategy designed to translate analytical insight into measurable operational disruption and lasting economic value for Warner Bros. Discovery.

Sports Piracy Landscape

Pirated live football content reaches viewers through a resilient, industrialized ecosystem fed by a shared upstream supply chain where origin servers re-encode the broadcast and CDNs distribute it at scale to three consumer product tiers.



Free Web Streams allow users to view live matches directly within a standard browser at no financial cost, generating the highest volume of traffic with single domains like futbollibretv.su attracting up to 8.8 million monthly visits in April 2026. These platforms act as lightweight frontends that embed external streams rather than hosting video themselves, meaning removing a single domain has minimal impact, while operators monetize via advertising, betting affiliates, and malicious software.

IPTV Subscription Applications like Magis TV and Xuper TV function as paid products that control access server side via user credentials, updating automatically outside traditional app stores to bypass marketplace bans. IPTV represents the most direct threat to premium revenue: blocking these services in Argentina in February 2026 triggered **31,000** incremental HBO Max subscriptions and a 67% surge in gross subscriber additions. The low barrier to entry of roughly USD 2,000 fuels a long tail of small operators and rogue cable providers that offer IPTV subscriptions together with ISP service.

Preloaded **TV Boxes** are physical Android or modified streaming devices sold with piracy applications preinstalled, offering a simplified user experience that is entirely immune to standard online DMCA takedowns. Because traditional monitoring systems track web

traffic rather than domestic hardware deployments, TV boxes represent a major operational blind spot where WBD lacks a reliable estimate of the total installed base.

Case analysis: the Zapping success case.

To protect its **exclusive broadcasting rights** for the LigaPro Ecuabet, the streaming company Zapping deployed a proprietary algorithmic system known as "Tío Emilio". This tool represents a proactive shift in the industry, moving away from slow legal takedowns and massive, disruptive network blocks. Instead, it establishes a new regional standard by utilizing a purely technological approach to track and neutralize illegal retransmissions during live events.

The algorithm of “Tio Emilio” traces dynamic metadata and imperceptible digital watermarks to attack the original extraction of the signal, known as source capture. This targeted method neutralizes the specific compromised session without affecting the general internet traffic or the high-quality viewing experience of legitimate subscribers.

A **key difference** that allows this anti-piracy system to function so efficiently is that **Zapping only goes through streaming** with no signal in traditional channels. Because the platform operates exclusively over an internet-based infrastructure rather than relying on legacy satellite or cable television networks, it maintains absolute end-to-end control over its video distribution. This purely digital environment enables the rapid deployment of software-native solutions, allowing Zapping to instantly revoke access tokens for pirated sessions directly at the edge of the network without depending on the sluggish interventions of traditional telecommunication providers.

A solution with this or a similar technology could be evaluated for HBO Max exclusive transmissions, since HBO Max relies entirely on streaming with no signal in traditional channels, WBD could embed dynamic session watermarking directly into its digital pipeline, similar to the “Tio Emilio” solution.

Warner Discovery Assessment

The current anti-piracy operational structure for Warner Bros. Discovery (WBD) in Latin America, specifically targeting high-exposure football rights in Chile and Argentina, is capable but fragmented, heavily manual, and lacking a centralized data layer. Anti-piracy responsibilities are treated as secondary duties across multiple disconnected teams, and the external vendor ecosystem operates without integration.

Internal Stakeholders & Operational Constraints

WBD's anti-piracy efforts are divided across four internal functions, none of which, except an unformalized delivery team, counts anti-piracy as its primary mandate.

- **Anti-Piracy Management (Delivery Team)**: A lean, two-person team that serves as the operational hub for Argentina and Chile. They consolidate vendor reports and coordinate escalations, though the team structure is not yet formalized.
- **Legal (Panregional & Local)**: Manages litigation, blocking orders, and industry memberships like MPA and Alianza. However, the critical protect@ escalation mailbox acts as an operational "black box": it lacks a ticketing system, case numbers, or automated status updates, forcing teams to follow up manually. Also, its action is very limited in countries like Argentina and Chile, where the official departments are very limited.
- **Digital Events Operations**: Responsible for live transmission health, meaning anti-piracy is a side job. While they successfully identified major infrastructure vulnerabilities, such as the Telecom Argentina (Flow) CDN leak, signal-side defense has hit diminishing returns and cannot stop restream-based piracy of HBO Max exclusives.
- **Social Media (TNT Sports Chile):** An editorial team of journalists doing manual comment-thread monitoring on official accounts during match days. Monitoring is entirely restricted to WBD's own pages, leaving high-impunity spaces like Reddit, Discord, and Telegram unmonitored.
- **Distribution Chile/Argentina & Finances:** regarding piracy, they are focused only on seeing the impact of piracy operations on Warner's business. They see a big lack of information regarding IPTV services and really dimensioning the impact of this in the business.

External Provider Matrix

WBD employs a mix of external vendors that do not integrate with one another, resulting in overlapping data and a fragmented understanding of the true piracy footprint.

Provider	Core Function	Strengths	Key Weaknesses & Limitations
IP House *(MarkScan)*	Act: Live detection and automated DMCA or delisting enforcements.	Weekly operational cadence: broad channel coverage across web, social, Telegram, and IPTV.	Unstable reporting metrics: inflated compliance figures where real video source compliance is ~30% vs. the headline ~81%, and blind to HBO Max exclusive feeds.
AF&M	Dissect: Near forensic infrastructure and app reverse engineering.	Deep technical visibility: maps the full server stack across K1, K2, and K3 layers to uncover real origin IPs outside Cloudflare.	Heavily labor intensive and per target: does not discover new sources, and outputs are stalled by slow local legal blocking frameworks.
Randy Labs	Find: SaaS automated discovery platform.	Real time automated discovery: high quality evidence capture and an API first architecture that allows data to push directly into internal tools.	Demo mode only: WBD has no formal contract or active account. Scope is currently limited to web and IPTV.
Alianza SSLA	Size: Monthly regional audience ranking and market sizing.	Sizes piracy by actual audience reach: excellent for long term strategic prioritization.	No live enforcement capability: relies on directional SimilarWeb estimates and misses live match day traffic spikes.

Friend MTS	Act: Event-based live internet monitoring, and copyright infringement enforcement.	Uses fingerprint matching to verify piracy against source content; directly integrates with Facebook Rights Manager (FBRM) and YouTube CMS for automated social media blocks; tracks upstream infrastructure/ASNs; and maintains a panel of covert IPTV subscriptions.	Slow live enforcement, as ~60% of infringing streams remain active for 60–120 minutes during critical live windows; Active for Argentina LPF, discontinued in Chile.
Fabric (BB Media)	Regional market intelligence mapping piracy supply & demand	Tracks LATAM user habits (m3u, IPTV); raw data access and social media trends.	Quarterly latency and no live enforcement, dedicated to do analysis only.

Because these external platforms are entirely disconnected, WBD is left with overlapping data, unstable metrics, and no single source of truth to guide live enforcement.

Internal Initiatives & Strategic Wins

Despite limited resources, internal teams have executed highly capable experiments and tactical wins

The Discord PPV Case: A Blueprint for Deterrence

The Chilean Social Media team carries out a manual monitoring of official TNT Sports accounts, as described in the following diagram.



In May 2026, while carrying out this monitoring, the Social Media team detected pirate resellers using Instagram to market illicit, paid access to exclusive HBO Max and TNT Sports matches via Discord.

- **The Operation:** By escalating the case to protect@, the offending Instagram accounts were terminated within hours.
- **The Breakthrough**: Because the resellers required proof-of-payment bank transfers, the team inadvertently captured the targets' full names, national IDs, and tax data.
- **Strategic Impact**: WBD passed this data to local Chilean counsel to file criminal complaints. The goal is to establish high-profile, exemplary cases that deter the broader reseller network rather than scaling endless enforcement resources against minor actors.

Internal Tooling (Morpheus & Compliance Tracking)

- **Project Morpheus:** Developed by the Operations team to automatically scrape Legal-provided lists of pirate sites and extract the origin CDNs serving the stolen streams. It has since been discontinued due to resource constraints.
- **Chile ISP Compliance Tool:** An active tool that exists through local ISP networks to verify whether providers are actively complying with judicially mandated blocking orders, such as Magis TV.
- **Core Limitation***:* Both tools function as list-driven validators of known targets rather than dynamic discovery engines for unknown infrastructure.

Diagnosis

WBD weak points

The discovery process identified three structural pain points that run across WBD's entire anti-piracy operation. These vulnerabilities are not isolated to any single team or vendor; instead, they compound each other and collectively explain why current efforts are less effective than the sum of their parts.

Data Fragmentation & Measurement Blindness

- **Siloed Vendor Formats**: Internal data feeds and external vendor outputs, including IP House workbooks, AF&M forensic reports, Randy Labs case data, and ALIANZA-SSLA rankings, arrive in mismatched formats and on disconnected schedules. Because they are consumed independently, there is no centralized data layer to aggregate them into a common structure.
- **The Measurement Blind Spot:** No single function within the organization holds a complete, real-time picture of the active piracy surface. Because different vendors detect different sources, total observed infringements vary wildly depending on which vendor database is queried.
- **Lack of Performance Baselines**: Without a reliable estimate of the true piracy universe, WBD lacks a consistent baseline to evaluate whether enforcement countermeasures are working, meaning operational progress cannot be conclusively demonstrated or validated with data.

Excessive Manual Work Across the Workflow

Every stage of the current operational workflow depends heavily on human intervention, with zero automated processes and no shared tracking infrastructure:

- **Fragmented Intakes**: The Social Media team must monitor official account comment threads entirely by hand, reporting infringing links informally via WhatsApp.
- **Manual Operational Handling**: The Anti-Piracy Management team acts as a manual clearinghouse, consolidating these fragmented inputs to escalate them via standard email to the protect@ mailbox.
- **The Mailbox "Black Box":** The protect@ intake system lacks automated ticketing, case numbers, or status updates. Simply tracking the lifecycle of an ongoing escalation requires sending manual follow-up emails.
- **Labor-Intensive Evidence Gathering**: Investigating a newly detected pirate source, including creating test accounts, logging in, navigating the platform, and capturing screenshots as legal evidence, is executed completely by hand for every single case.

- **Concentrated Risk & Capped Throughput**: Interpreting and cross-referencing vendor reports requires specific, non-systemized expertise that sits with a very small number of individuals. This concentrates operational knowledge in people rather than systems, forcing the team to spend their limited time reading reports and gathering evidence instead of taking strategic action.

Slow & Structurally Limited Official Enforcement

- **The Live Sports Time Gap**: The average turnaround time for a standard DMCA takedown is approximately 59 minutes. For a time-sensitive 90-minute live fixture, this means premium broadcast content remains fully exposed for at least an entire half before source-level mitigation takes effect.
- **Superficial Countermeasures**: Basic Google delisting reduces organic discoverability but fails to actually remove or stop the underlying live stream.
- **Enforcement Infrastructure Deadlocks**: Real-time dynamic domain blocking, the only effective mechanism to counter pirates as they rotate domains mid-match, is completely unavailable in Chile and Argentina. This is due to the lack of three mandatory operational conditions: a live detection feed, a direct technical interface to local ISPs, and a blanket legal authority covering new domains without per-filing mandates.
- **IPTV Structural Resilience**: Standard takedown workflows only successfully target app download links and online tutorials. Active subscribers who have already installed piracy applications remain entirely unaffected, allowing the illicit services to continue functioning normally for their existing user base regardless of distribution-channel enforcement.

Pirate weak points

The same Discovery process identified **structural weaknesses** in the pirate ecosystem itself. These are points where the operating model of each service type is exposed to disruption, not through the official action channels described above, but through targeting the dependencies the pirate services rely on.

For **web streams**, the business model is advertising-dependent. Without ad revenue, the economic case for operating these sites deteriorates. The infrastructure also struggles under high concurrent load: these sites are optimized for typical organic traffic, not for traffic spikes, which creates a potential avenue for degrading user experience during the windows that matter most.

For IPTV and subscription-based services, revenue collection is conducted entirely through mainstream digital payment infrastructure like Mercado Pago, bank transfers, and credit cards. These channels are regulated, traceable, and subject to takedown or

freezing. Separately, subscriber acquisition is mediated through messaging platforms such as WhatsApp, Telegram, and Instagram direct messages. Because human operators manage these flows, there is a throughput ceiling: the reseller network can only process a limited number of new subscriptions at any given time. This is an operational bottleneck that can be targeted.

Hypothesis: Two user profiles

Based on our research of the field and the meetings we held with the different stakeholders, pirate sports consumption in LATAM does not behave as a single audience. It splits into two distinct user profiles with different motivations, expectations, and, most importantly for us, very different odds of ever converting into paying customer

The web stream user. The audience behind free, browser-based sites such as Fútbol Libre or Pelota Libre.

- In raw numbers, this is the **larger and more visible group**: web streams aggregate the most traffic and the most visitors, and it's where we have the clearest read on volume.
- Their behavior is shallow and occasional: someone opens a single match because they care about that one game, then leaves.
- They tolerate poor resolution, unstable feeds, and the malware and advertising that come with these sites.
- This is essentially **a pirate by nature**: the appeal is that it's free, and the user is very unlikely to change how they consume no matter what we do.

The IPTV user. A fundamentally different profile, closer to a traditional pay-TV subscriber than to a casual streamer.

- Services like MagisTV (recently rebranded Súper), Flujo TV, and others are consumed the way a pay-TV or premium OTT product is: through a set-top box or app, with 1,000+ channels, linear plus on-demand content, an electronic guide, and an operator-grade experience.
- This user does not dip in for one match; they rely on the service for everything like sports, series, and film alike.
- Critically, **they already pay a subscription**: one MagisTV-type package can cost roughly a third of an official subscription while bundling the equivalent of four or five legitimate services.
- The **scale is significant**: partners estimate that around 20% of the broadband network is used for this kind of content, which against roughly 4.8M broadband households implies on the order of a million potential IPTV subscriptions in Chile alone.

- ISPs share the concern, given how much bandwidth these services consume relative to web streams.

Between the two, web streams are the larger and more measurable source, but **we believe the IPTV user is the better target**:

1. Breadth of use raises the stakes of disruption. Because the **IPTV user depends on the service** for all of their content rather than the occasional match, losing it is a genuine disruption giving them a far stronger reason to seek a legitimate replacement.
2. They have already demonstrated a **willingness to pay**. The IPTV model is subscription-based, so this user has crossed the threshold of paying for content; the distance to an official subscription is one of price and awareness, not of willingness. Many of these users may not even recognize the service as illegal precisely because they pay for it — which opens an education and enforcement lever that simply does not exist for the free-stream audience.
3. It gives us a clearer read on impact. Targeting IPTV lets us build a sharper picture of the traffic these services actually carry and observe how a disruption affects the end-user experience, a feedback loop the diffuse web stream audience does not offer.

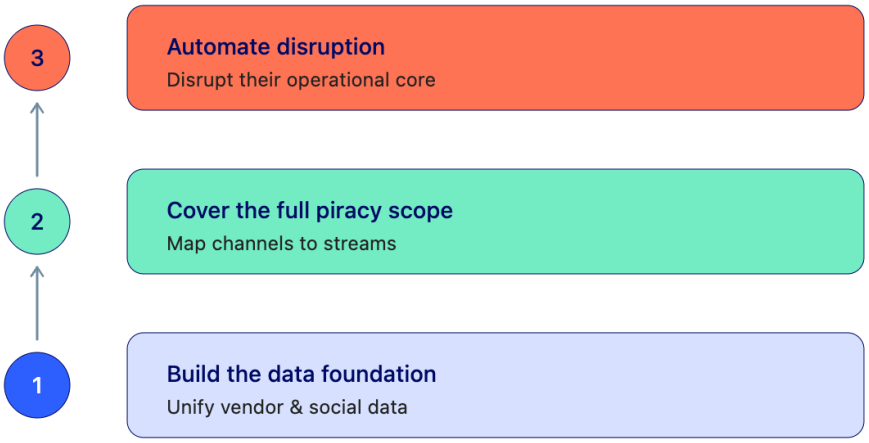
In short, the hypothesis is a deliberate trade-off: we think that we should prioritize the higher-value, more convertible segment over the larger but shallower one. The web stream user is unlikely to ever pay; the IPTV user already does.

WBD Antipiracy Implementation Path

Based on the previous diagnosis, we outline an implementation plan that points into solving the different pain points in WBD against piracy, mainly for sports transmissions, but reusable in other scopes. As was seen in the previous section, the problem is complex enough to require an end to end approach that treats the pirate ecosystem as a whole instead of only targeting specific cases one by one.

Three layer implementation plan

To systematically address Warner Bros. Discovery's structural vulnerabilities, specifically fragmentation, heavy manual dependency, and delayed enforcement, this implementation path points to building a modular, three-layered framework. Rather than treating detection and enforcement as isolated incidents, this proposed path **outlines an interconnected ecosystem** where intelligence can eventually fuel automated counter-operations. These layers are:



The strategy begins with **Layer 1: Building the Internal Data Foundation**, which points to creating a centralized operational hub inside WBD. This foundational layer is designed to provide a single point of truth accessible to stakeholders across the organization, from business leadership to the legal department, unifying both insights and future actions.

The **immediate benefit** of this layer is unlocking the rich, underutilized data from premium vendors WBD already invests in, such as IP House, FriendMTS, and RandyLabs. Currently, these valuable insights are lost in siloed email attachments and disconnected spreadsheets. By automatically ingesting and standardizing these streams, the first layer ensures that existing vendor intelligence is maximized and transformed into a shared corporate asset.

Then, the **second layer: Covering the full piracy scope**, which can be built simultaneously, points to covering the full piracy scope by **illuminating the critical blind spots** in the current regional ecosystem. This phase goes beyond superficial stream detection to map the entire lifecycle of pirate operations, uncovering how illicit IPTV applications are distributed, how they leverage social media for audience acquisition, and how they exploit digital payment infrastructure. Instead of only recording isolated site names with screenshots as evidence (which is what current vendors do), this second layer is designed to construct an intelligence map that connects the dots from initial social media promotion down to the actual underlying video transmission. This deep visibility allows WBD to map multiple rogue social media accounts back to the exact same pirate source.

Furthermore, this layer can be modularly expanded to police other ecosystems beyond social media. For example, it could audit web-stream advertisements to identify if any legitimate WBD advertising partners are inadvertently funding pirate platforms, giving our legal team immediate leverage to issue high-impact compliance complaints.

Finally, as these data and intelligence streams converge, the framework culminates in **Layer 3: Automating Actions to Disrupt Pirate Sources Operations**, which points to building the capabilities needed to degrade the economic and operational stability of the pirate services themselves. Because traditional enforcement suffers from crippling live-sports time gaps, this final tier shifts the focus toward active operational disruption.

By understanding the specific operational friction points mapped in the second layer, this tier outlines targeted countermeasures designed to disturb digital payment gateways, disrupt customer communication channels, and cap subscription processing throughput. Working in unison, these three layers lay out a seamless, highly scalable loop to autonomously navigate the piracy ecosystem, gather ironclad evidence for the WBD Legal team, and inflict real operational costs on illicit networks.

Operational & Economic Benefits

As mentioned before, in Chile alone, **WBD subscriptions** to live football have fallen from roughly 1M to 600K since 2021, with much of that loss absorbed by pirate services. On top of this, an estimated 1M households now rely on a pirate IPTV service as their primary streaming service. At roughly USD 16 per subscription per month, winning back just 10% of those households (100,000 subscribers) would represent USD 1.6M in monthly revenue in Chile alone, and the same pattern extends to Argentina and the rest of LATAM. A recoverable base of this scale justifies building a stronger structure to tackle the problem.

Faced with an ecosystem this complex and resilient, the defining strength of this three-layered path is its shift from chasing individual, isolated live streams to targeting the piracy system as a whole. Rather than fighting game by game with automated DMCA takedowns on specific streaming links, which vendors already do, this approach maps the entire infrastructure behind illicit providers.

This lets us strike at their operational core rather than a single live stream:

- Standard external providers focus on automated DMCA takedowns of specific streaming links during a live match, which fails to impact the broader infrastructure or active IPTV subscribers.
- By **targeting an entire IPTV service** fully rather than just an isolated match feed, this path inflicts damage on core operational vulnerabilities like regulated digital payment channels and communication channels, affecting the user experience, which can lead to users returning to official subscriptions.

- For webstreams, we can also target its revenue model, doing a registry of the companies that use advertising spaces on pirate services.

As seen in the Valoroso Operation, successfully disrupting an entire IPTV network at a systemic level can directly trigger tens of thousands of incremental official subscriptions and sharply increase gross subscriber additions.

On the operational side, this approach replaces fragmented monitoring with a full-scope map of the piracy ecosystem. It draws on the intelligence that current paid vendors already provide but that today sits siloed across separate tools and email chains, and complements it with **active social media monitoring**. The result is that we no longer track isolated URLs but complete chains that run from the initial social media promotion all the way to the live stream. Today, getting this information involves sending up to 4 emails to multiple stakeholders, requiring a couple days to get the full picture (if possible).

This **system-wide view** fundamentally changes how WBD uses its existing vendor ecosystem. Today, premium telemetry from providers like IP House, AF&M, and Randy Labs arrives in mismatched formats, fragmented pieces of the puzzle with no centralized view of the true piracy footprint. The proposed framework does not replace these vendors; it **maximizes their value** by aggregating their data into a single intelligent layer that surfaces the hidden connections between seemingly unrelated pirate sites.

By automating evidence collection and cross-referencing incoming streams to map the full network, this data foundation **removes the heavy manual workload** that currently bottlenecks internal teams. For example, currently, the Social Media team in TNT Sports Chile manually monitors the official social media for pirate sources, takes screenshots manually and sends the evidence to the Anti Piracy Management team via Whats App. All these steps can be automated with the Layer 2 developments.

The operational benefits are substantial:

- Smarter use of existing paid-vendor reports
- Less manual work, consolidated into a single source of truth for streaming piracy
- End-to-end mapping of pirate systems, from distribution channels to payment

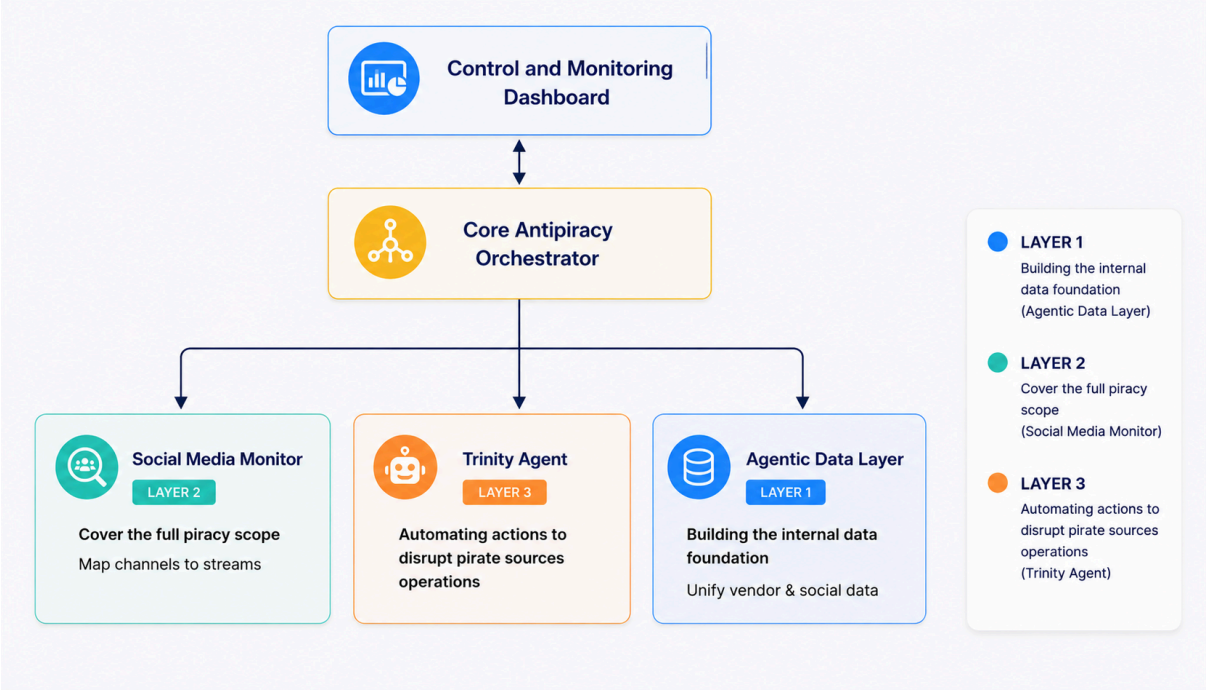
With this intelligence in hand, further strategies can be developed, beyond the ones above, to win back pirate subscribers. For instance, the same distribution channels the pirates rely on could be seeded with official WBD subscription promotions, inserting WBD directly into the middle of their operations.

The end result is twofold: WBD gains both a full picture of the pirate ecosystem and the means to strike at its operational core.

PoCs Design - Short Term Roadmap

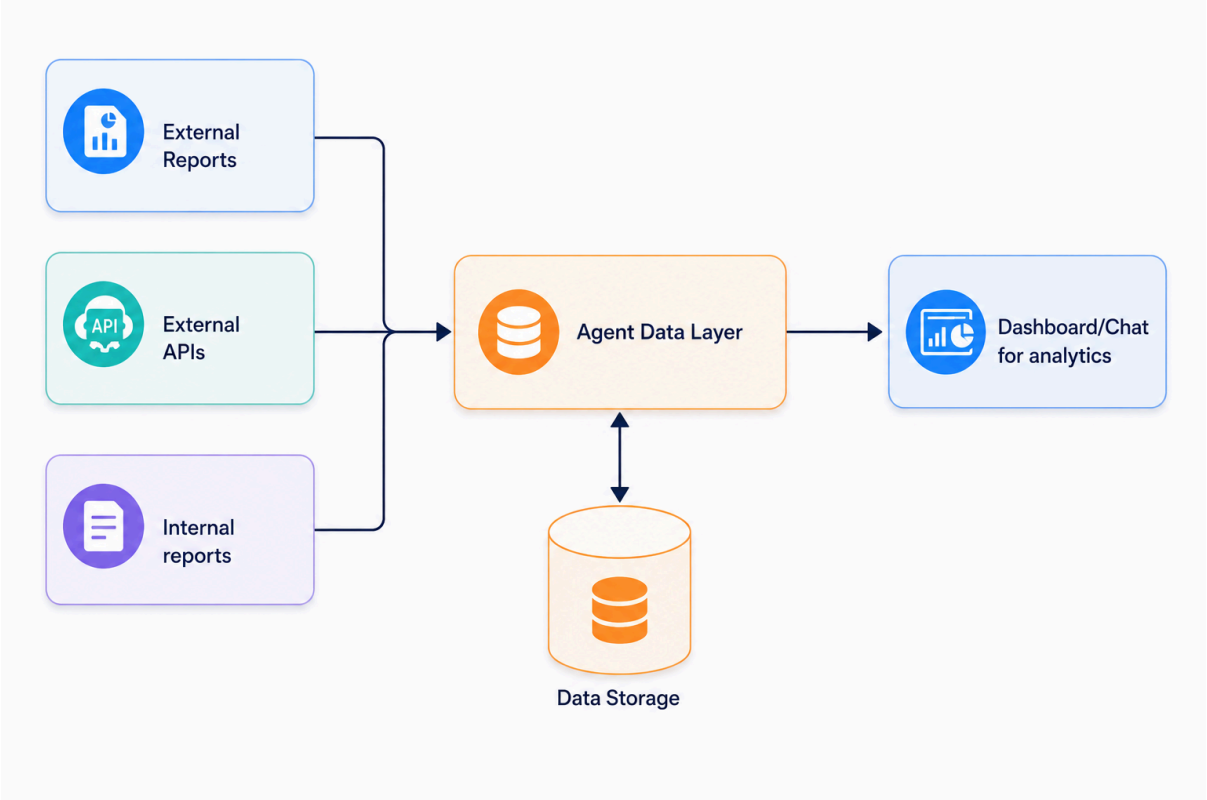
The following **Proofs of Concept (POCs)** were designed as the foundational building blocks of WBD's long-term anti-piracy transformation roadmap. Rather than approaching piracy through isolated, market-specific tools, these initial developments are intentionally structured as scalable components of a unified ecosystem. Each PoC targets a critical operational layer (data consolidation, social media intelligence, and active disruption) while sharing a common architectural philosophy centered on modularity, adaptability, and progressive automation. Although the initial implementations focus on validating high-impact use cases within a controlled scope, the broader objective is to **evolve these capabilities into a fully integrated**, agent-driven anti-piracy platform capable of adapting to the continuously changing piracy landscape.

As it can be seen in the diagram below, the three POCs were designed to be generalistic and with the target of integrating them into a single ecosystem. This targets two specific goals: that the developments are not market specific and that they can adapt to the evolution of the piracy ecosystem.



In the following sections, for each Agent, we described the objectives, success criteria and high level technical design.

Data Layer Agent



Objective

This POC targets Layer 1 of the implementation plan, transforming Warner's anti-piracy operations by directly **solving data sparsity** and **eliminating the friction** of conflicting vendor reports. By automatically ingesting and standardizing fragmented streams from IP House, AF&M, RandyLabs, and internal pipelines into a single source of truth, it replaces manual report analysis with a centralized dashboard and a conversational, natural-language interface for instant insights.

Beyond simplifying workflows and organizing legal requests, the agent shifts WBD **from a reactive** posture **to active**, real-time threat monitoring. Crucially, it provides the business with a definitive way to measure and compare the effectiveness of its actions, whether driven by internal teams, the PoCs agents, or external countermeasures like dynamic DNS blocking.

In the following image, it can be seen an example of how the information can be structured using a knowlegde graph, creating a clear view

Benefits

As detailed in the previous section, WBD's anti-piracy intelligence is heavily bottlenecked by fragmented, manual workflows, with critical information flow scattered across disparate email threads, Slack channels, and WhatsApp messages.

To illustrate the friction within the current operational setup:

- **Siloed Knowledge**: Accessing a standard vendor report (such as an IP House update on Chilean football) requires emailing an overextended Anti-Piracy Management team member, for whom anti-piracy is not a sole responsibility, waiting for a manual search, and relying on them to interpret the raw results.
- **Manual Normalization**: Building a comprehensive situational assessment across Argentina and Chile forces stakeholders to manually request, parse, and cross-reference completely mismatched report formats and categories from Alianza, IP House, Randy Labs, and AF&M.
- **Evidence Gathering**: if someone wanted to gather all the evidence existent for a given site, would have to go report by report, manually, across every game, to find all the information. And not only that, as it would not have a place to check if the site is really new or was already reported previously

By establishing a centralized, agentic data layer to automatically ingest and standardize these disparate inputs, WBD collapses these multi-step friction points into a single, cohesive command center. This transformation delivers five core benefits:

- **Unified Threat Monitoring**: Consolidates active pirate services, tracking their traffic metrics, illicit advertising channels, and forensic evidence in a single dashboard.
- **Automated Cross-Reference & Detection**: Automatically cross-checks incoming data across multiple vendor streams and internal discovery agents to instantly flag newly emerging pirate sources.
- **Dynamic API Integration**: Connects seamlessly with external platforms like Randy Labs to automatically pull and enrich existing profiles with real-time telemetry.
- **Natural-Language Insights**: Eliminates tedious spreadsheet analysis by allowing stakeholders to query the database using plain language, returning accurate, data-driven answers in seconds.
- **Autonomous Escalation Workflows**: Streamlines legal and operational reporting by utilizing the agent to autonomously generate formatted escalation emails for protect@wbd.com.

The strategic impact of implementing this Data Layer Agent directly addresses the overlapping data and unstable metrics noted in the WBD assessment. It establishes a definitive, single source of truth, giving WBD total visibility into where it stands in the anti-piracy fight and empowering all involved internal functions to execute high-return countermeasures.

Success Criteria and KPIs (WIP)

Success for this POC is measured across three dimensions: operational efficiency,

financial impact, and threat containment. The following KPIs establish a before/after baseline that business stakeholders can track directly.

Operational Efficiency

- **Report ingestion rate:** ≥90% of target reports (IP House, AF&M, RandyLabs) ingested and normalized without manual intervention, replacing an estimated X hrs/week of analyst work.
- **Time-to-insight:** Query resolution reduced from X hours (manual) to minutes.
- **Workflow consolidation:** Steps required for a full cross-vendor situational assessment reduced from X steps to 1 (single dashboard query).

Financial Impact

- **FTE cost avoidance:** Estimated $X/month recovered by eliminating manual report parsing, calculated as (hours saved/week × blended hourly rate).
- **Legal escalation speed:** Time to generate a formatted escalation to protect@wbd.com, shrinking the revenue leakage window per active pirate source.
- **Evidence consolidation:** % of escalation cases where all supporting evidence was automatically aggregated vs. manually collected across reports.

Baseline measurement protocol: Prior to go-live, WBD and Marvik AI will jointly record current-state values for each metric above. Post-implementation measurement at 30 and 60 days will quantify the delta and validate ROI.

Technical Design

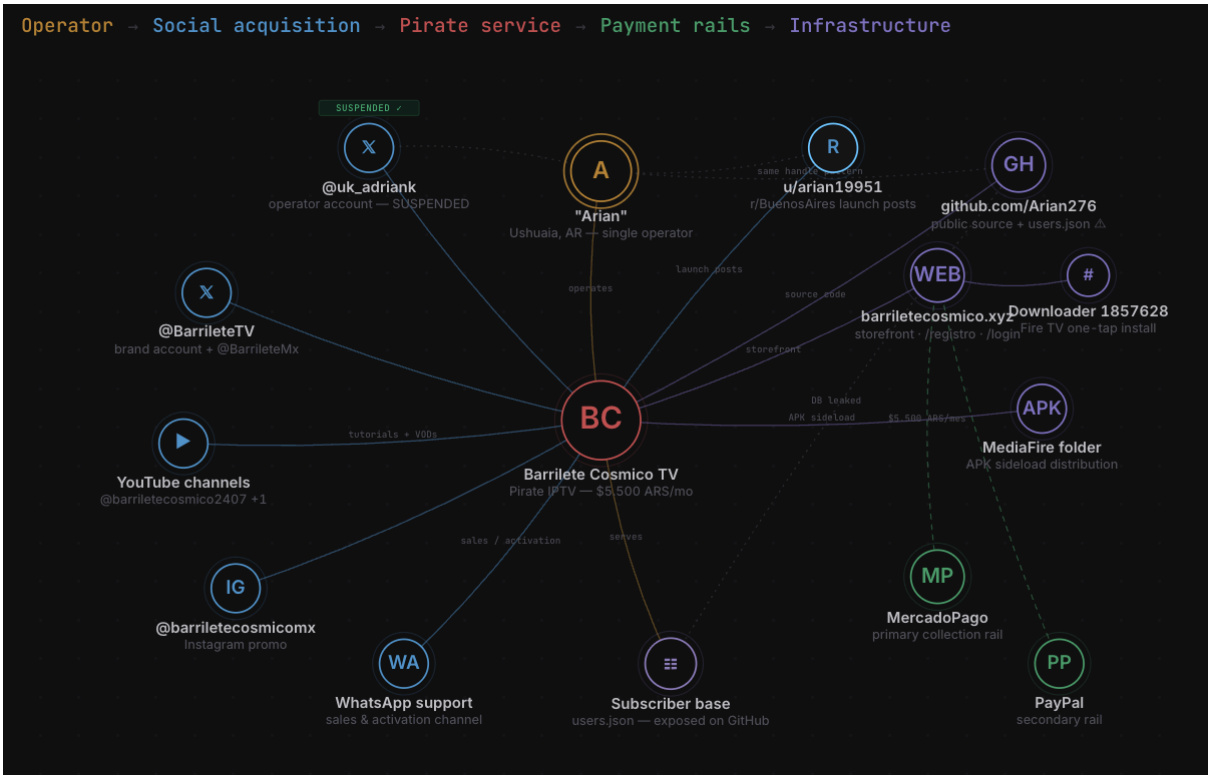
The Data Layer POC will be built as an agentic system designed to **automatically ingest** fragmented vendor intelligence and expose it through a unified, conversational interface.

Structurally, it follows a modular **ingestion-to-query** pipeline, divided in three layers:

- **Centralized Storage and Registry**: reports like those from IP House, AF&M, Randy Labs, and ALIANZA-SSLA are stored and versioned in a single storage system, simplifying its access and usage.
- **Information processing**: regardless of format, each document is parsed, semantically chunked, and normalized into a single source of truth with standardized format.

- **Agentic Graph Generation**: to be able to understand the full scope of each pirate service, Knowledge Graphs are built to cover the full spectrum of the pirate service, from diffusion media to revenue streams.

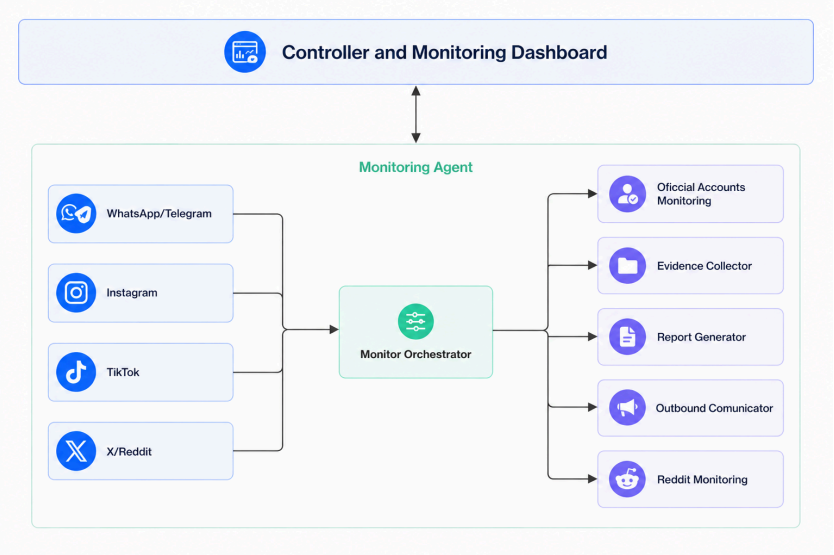
In the following image, it can be seen as an example of how a knowledge graph can be built to showcase the full spectrum of a given pirate service, from social media accounts to URLs and payment methods.



Knowledge Graph example for Barrilete Cosmico TV

Beyond ingestion, the system exposes a natural-language interface that allows stakeholders to query the consolidated report base directly. The underlying agent applies a corrective retrieval loop, searching, grading result relevance, and reformulating queries when needed, before generating grounded, source-cited answers. A dashboard layer connects to this same agent core to surface consolidated threat metrics and enable direct querying without spreadsheet analysis or manual cross-referencing across vendor reports.

Social Media Monitor Agent



Objective

This POC targets the **Layer 2** of the implementation plan, the primary goal is to achieve automated discovery and evidence-gathering, with the business goal of making it very hard for pirates to use social media for getting to the public.

The agent consumes information from external sources, like current vendor reports (IP House, AF&M) and uses that information to **explore the social media ecosystem**, from monitoring official TNT Sports accounts (currently done manually) to exploring common piracy sources like Facebook groups and Reddit.

Beyond monitoring, the agent has the autonomy to actively use social media. It can comment and send messages across social platforms, including WhatsApp, Telegram, Discord, and Instagram while taking natural-language tasking from the team to raise alerts for the Piracy Management team. Ultimately, it gathers evidence on paid pirate services, sends it to protect@, and leaves the case completely ready for the Trinity Agent to take action.

The business long term end goal of this development is making it more and more difficult for pirates to use social media for getting to the public.

Given the nature of a PoC operating on a limited scope of the full piracy ecosystem, success targets are split between short-term operational metrics and long-term business impact metrics.

Short-Term KPIs (POC Validation)

- **Detection volume uplift:** ≥5x increase in piracy-related publications detected weekly compared to the current manual social media monitoring baseline.

- **Piracy source identification rate:** Total number of unique pirate sources (accounts, groups, channels) surfaced per week, establishing a baseline for long-term trend tracking.
- **Detection precision:** % of flagged accounts/publications confirmed as true positives vs. false positives, target ≥80% precision to ensure enforcement resources are not wasted.
- **Evidence package completeness:** % of detected cases where the agent automatically produces a complete, escalation-ready evidence package for protect@wbd.com without manual enrichment.

Long-Term Business KPIs

- **Social media piracy surface reduction:** Month-over-month reduction in active publications sharing access methods to unauthorized TNT Sports streams — directly measures how much harder it becomes for pirates to reach the public via social media.
- **Takedown cycle time:** Reduction in average time from a pirate posting an enforcement-triggering link or invite to successful takedown — shorter cycles directly reduce the audience reached by each pirate post and minimize subscription displacement.
- **User-facing discoverability index:** Periodic measurement of how many working unauthorized stream links a standard user can find via basic keyword search on social platforms — reduction correlates directly with subscriber acquisition protection.

Technical Design

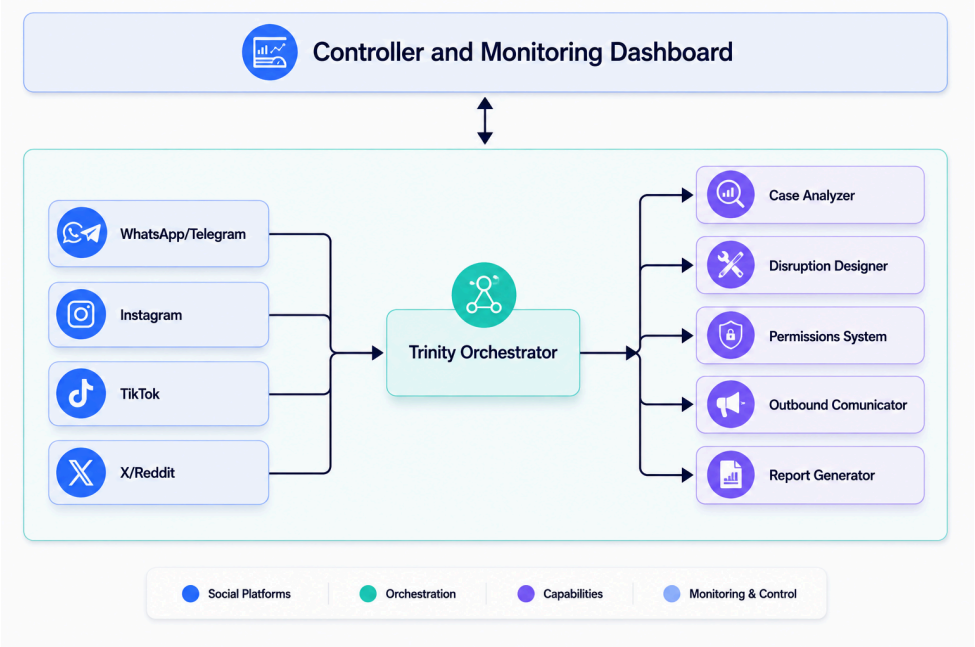
This system is built as an **OpenClaw-based agent** designed to consume existing data sources and automate the manual pipeline of evidence gathering and report generation. Structurally, it features a modular end-to-end design that allows for progressive integration and development. All communications connect directly to the same centralized agent core, making the entire setup easy to adapt from one market to another.

Beyond the Agent, the system involves the implementation of a front end that allows it to both manage the behavior of the agent, consume the information that obtains and track the ongoing tasks the agent is performing. For example, through this front end, the users can:

- Provide specific instructions to the agent with natural language. For example, a member from the team can target the agent to a specific Facebook Group/Reddit Thread
- The agent publishes cases that require human intervention for advanced steps and information gathering.

Finally, the agents connect with the Data Layer to publish all the gathered evidence into the central information source which matches it to previously detected sources.

Trinity Agent



Objective

This field agent is designed to act directly on prepared cases to **disrupt pirate operations**, focusing heavily on reducing slow enforcement timelines. It consumes information from the Data Layer and the Social Media Monitor to analyze infringement cases and design tailored disruption actions.

These actions specifically target the **operational weak points** of pirate services, such as their sales channels, advertising networks, and digital payment methods.

This agent represents the most complex of the three POCs, as it involves directly actioning against the piracy services. Because of this, instead of trying to come up with a generalistic agent, the first tests in this line of action will be specifically designing campaigns for a given pirate service, always with the end goal of generalization. This approach will allow us to see rapidly what works on a specific case and how we can scale that to more cases, going from specific to generalist agentic applications.

Short-Term KPIs (Per-Case Validation) (WIP)

- **Operational disruption signal:** For message-based operations, average pirate response time before vs. during/after disruption — a significant increase indicates successful operational pressure on that service.
- **Sales channel degradation:** For targeted services, measurable reduction in the number of active sales touchpoints (WhatsApp numbers, payment links, promo accounts) operational before vs. after a disruption campaign.
- **Time-to-disruption:** Time elapsed from case intake (from Data Layer or Social Media Monitor) to first confirmed disruptive action — target to be set per case type as playbooks mature.
- **Playbook reuse rate:** % of disruption tactics developed for a specific case that are successfully generalized and applied to subsequent cases — key indicator of the POC's path toward scalable, agentic enforcement.

Long-Term Business KPIs

- **Subscription uplift correlation:** Increase in new subscriptions or reduction in churn observed in markets where a disruption campaign is active — the clearest direct revenue tie for this POC.
- **Pirate service recurrence rate:** % of disrupted services that resume operations within 30/60/90 days — lower recurrence means sustained revenue protection, not just momentary disruption.
- **Cost-per-disruption:** Total operational cost (agent + human oversight) to take down one pirate service, tracked over time as the agent generalizes — target is a declining curve as playbooks scale.

Implementation Timeline & Methodology

The POCs implementation will focus first on building the **Data Layer Agent** first, as it will set the ground for the other two PoCs development. In the following timeline it can be seen the expected outcomes for the following weeks together with the long term continuous development.

The Marvik team is in charge of developing the first versions of the apps, working interactively with the WBD anti piracy team in short development loops, ensuring that the developments provide business value. The Marvik team will also be in charge of escalating the PoCs to production and maintaining them.

Week to Week deliverable.

In the following table you can see a detail of the week-to-week expected deliverables for the first 4 weeks of the engagement.

Week	Deliverable	Business Value
First	• Mockups design of final web interface. • Clear definition of the information that will be collected from piracy-related documents. • First examples of processed documents.	• Provides visibility into how information will be processed and the type of insights the platform will generate.
Second	A first working version of: • AI Agent that automatically analyzes documents and extract information. • Dashboard that shows processed information.	• Simplified visualization of the data extracted so far. • Allow us to iterate on the solution and detect data inconsistency.
Third	• First version of the Knowledge Graph generation agent. • First simple visual piracy maps.	• Shows how piracy services, websites, domains, and social media accounts are connected. • This first visualization will allow us to continue iterating on how to structure these piracy maps.
Fourth	• Integration of Knowledge Graphs into interactive web for analysis. • A single interface combining the dashboard and relationship maps.	• Will be a functional proof of concept demonstrating the platform's core capabilities. • Helps identify networks and relationships that are difficult to see individually. • Allows users to explore data, connections, and trends in one place.

Roadmap

The initiative will take place over a 4-week period, structured into weekly sprints that will conclude with the delivery of specific results at the end of each one.

Module	Weeks									
	1	2	3	4	5	6	8	9	10	...
V0 Data Layer - Doc Processor										
Data Layer - Graph Builder										
V0 Interactive Frontend										
V0 Social Media Agent										
Disruptions Experiments										
Deployment, Scaling and Maintenance										